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# AB Quest

(ADHYANT BRAIN QUEST)

For

NASCENT COURSE

Time : 2 Hrs.

(For VIII to IX Moving Students)

Maximum Marks : 320

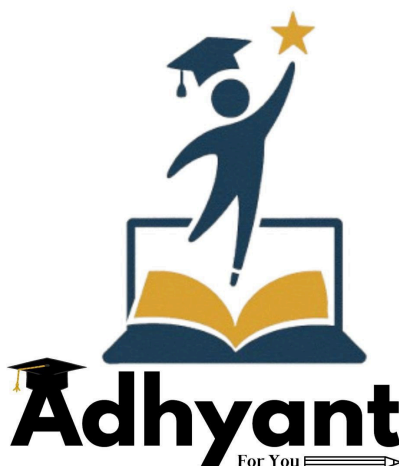
Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

Things NOT ALLOWED in EXAM HALL : Blank Paper, clipboard, log table, slide rule, calculator, camera, mobile and any electronic or electrical gadget. If you are carrying any of these then keep them at a place specified by invigilator at your own risk

## INSTRUCTIONS

1. This booklet is your Question Paper. **DO NOT** break seal of Booklet until the invigilator instructs to do so.
2. Fill your Form No. in the space provided on the top of this page.
3. The Answer Sheet is provided to you separately which is a machine readable Optical Response Sheet (ORS). You have to mark your answers in the ORS by darkening bubble, as per your answer choice, by using black & blue ball point pen.
4. Total Questions to be Attempted 80. Part-I : 20 Questions & Part-II : 60 Questions.
5. After breaking the Question Paper seal, check the following :
  - a. There are **11 pages** in the booklet containing question no. 1 to 80 under 2 Parts i.e. Part-I & Part-II.
  - b. Part-I contains total 20 questions of IQ (Mental Ability).
  - c. Part-II contains total 60 questions under 4 sections which are - Section (A) : Physics, Section (B) : Chemistry, Section(C) : Biology & Section (D): Mathematics.
6. Marking Scheme:
  - a. If darkened bubble is RIGHT answer : **4 Marks**.
  - b. If no bubble is darkened in any question: **No Mark**.
  - c. **Only for part - II** : If darkened bubble is WRONG answer: **-1 Mark (Minus One Mark)**.
7. Think wisely before darkening bubble as there is negative marking for wrong answer.
8. If you are found involved in cheating or disturbing others then your ORS will be cancelled.
9. Do not put any stain on ORS and hand it over back properly to the invigilator.

**NOTE :- Return this Test Paper**





**PART-I**

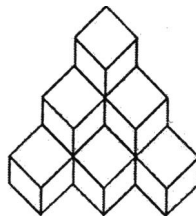
**IQ (MENTAL ABILITY)**

This section contains **20 Multiple Choice Questions**. Each question has four choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

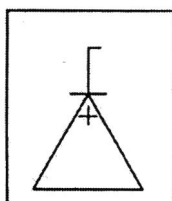
1. If 'P + Q' means 'P is the son of Q' and 'P = Q' means 'P is the sister of Q'. Then what does 'P = R + Q' means ?  
 (1) P is the father of Q (2) P is the daughter of Q  
 (3) P is the uncle of Q (4) P is the cousin of Q
2. In a certain code language, the word 'RECTANGLE' is coded as TGEVCPING, then how is the word 'RHOMBUS' coded ?  
 (1) TJOQDWV (2) UVWTJQN (3) TJQODWU (4) JTQOEWN
3. What is the number of triangles in figure ?



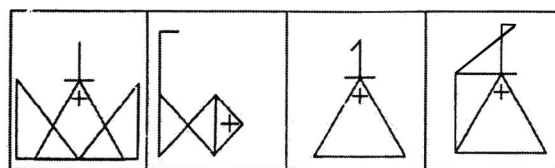
- (1) 16 (2) 28 (3) 32 (4) 38
4. Count the number of cubes in the given figure



- (1) 6 (2) 8 (3) 10 (4) 12
5. You go North, turn right, then right again and then go to the left. In which direction are you now ?  
 (1) North (2) South (3) East (4) West
  6. In the following question, you are given a figure (X) followed by four alternative figures (1), (2), (3) and (4) such that fig. (X) is embedded in one of them. Trace out the alternative figure which contains fig. (X) as its part.



(X)



(1)

(2)

(3)

(4)

7. Find the missing number

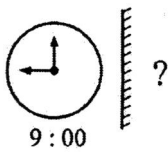
|   |     |     |
|---|-----|-----|
| 1 | 7   | 9   |
| 2 | 14  | ?   |
| 3 | 105 | 117 |

- (1) 16 (2) 17 (3) 12 (4) 18

8. If + means  $\div$ ,  $\times$  means  $-$ ,  $\div$  means  $\times$  and  $-$  means  $+$ , then  $8 + 6 \times 4 \div 3 - 4 =$

- (1) -12 (2) -20/3 (3) 12 (4) 20/3

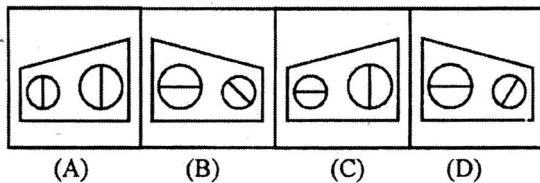
9. Find the mirror image



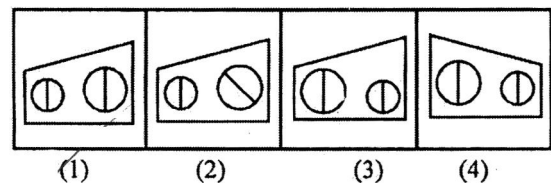
- (1) 12 : 00 (2) 5 : 00 (3) 3 : 00 (4) 6 : 00

10. Study the problem figures marked (A), (B), (C) and (D) carefully and try to establish the relationship between them. From the answer figures marked 1, 2, 3 and 4. Pick out the figure which most appropriately completes the series.

**Problem Figures**



**Answer Figures**



11. In a row of 30 boys, R is fourth from the right end and W is tenth from the left end. How many boys are there between R and W ?

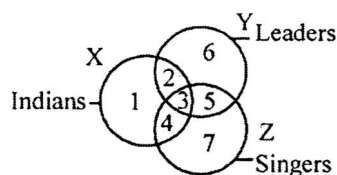
- (1) 15 (2) 16 (3) 17 (4) 13

12. Choose the missing term from the given options

1, 1, 3, 9, 6, 36, 10, 100, .....225

- (1) 15 (2) 16 (3) 20 (4) 25

13. Study the following figure carefully and answer the given question



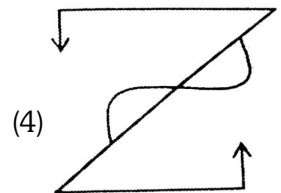
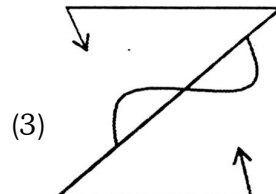
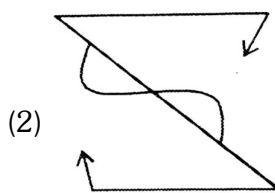
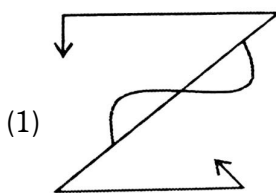
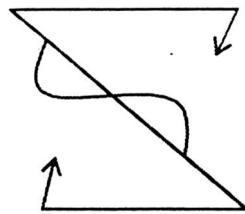
Which region denotes Indian leaders who are not singers ?

- (1) 2 (2) 3 (3) 4 (4) 5

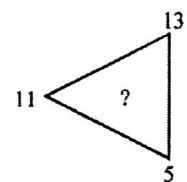
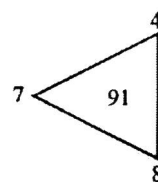
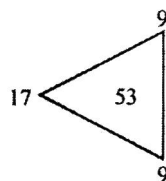
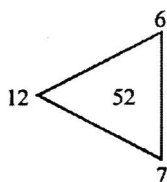
**14.** Find the correct water image for the following questions

STORE  
----- ← water surface  
?

- (1) SLORE                      (2) SLORE                      (3) SLOBE                      (4) SLOBE
- 15.** Neelam, who is Rohit's daughter, says to Indu, 'your mother Reeta is the younger sister of my father, who is the third child of Sohanji.' How is Sohanji related to Indu ?
- (1) Father-in-law                      (2) Father                      (3) Maternal-uncle                      (4) Grandfather
- 16.** In a code language, if TRAINS is coded as RTIASN, how will FLOWER be coded in the same language?
- (1) LFLOWER                      (2) LFWORE                      (3) WORELF                      (4) ERFLOW
- 17.** A person starts from a point A and travels 3 km eastwards to B and then turns left and travels thrice that distance to reach C. He again turns left and travels five times the distance he covered between A and B and reaches his destination D. The shortest distance between the starting point and the destination is :
- (1) 12 km                      (2) 15 km                      (3) 16 km                      (4) 18 km
- 18.** Select the mirror image of the figure when mirror is placed on the top of the figure ?



- 19.** How many numbers from 1 to 100 are there each of which is not only exactly divisible by 4 but also has 4 as a digit?  
(1) 7 (2) 10 (3) 20 (4) 21
- 20.** What number should replace the question mark ?

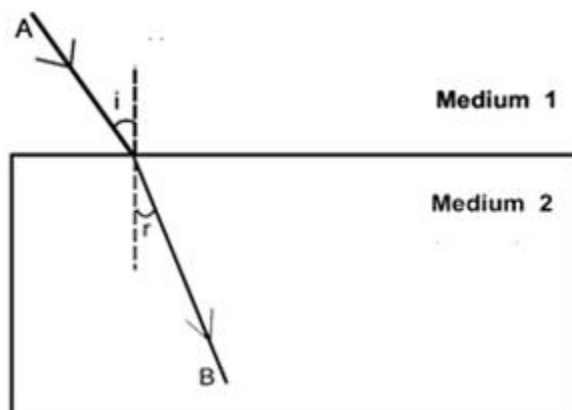


- (1) 89                      (2) 132                      (3) 92                      (4) 153

**PART-II**  
**SECTION-A : PHYSICS**

This section contains **15 Multiple Choice Questions**. Each question has four choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

- 21.** During electrolysis of an electrolytic solution, identify correct reactions set at anode and cathode.  
 (1) Anode - Oxidation, Cathode - Reduction      (2) Anode - Oxidation, Cathode - Oxidation  
 (3) Anode - Reduction, Cathode - Oxidation      (4) Anode - Reduction, Cathode - Reduction
- 22.** When a ray of light goes from one medium to another medium as shown, then its



- (1) Speed increases and wavelength decreases      (2) Speed decreases and wavelength increases  
 (3) Speed decreases and wavelength decreases      (4) Speed increases and wavelength increases
- 23.** In which of the following case, friction is NOT desirable?  
 (1) Writing in copy      (2) Turning a car on level road  
 (3) Car parked on a slope      (4) Motion of satellite around earth
- 24.** The class of force different from other is  
 (1) Pulling a cart      (2) Stretching a coiled spring  
 (3) Kicking a football      (4) Force of repulsion between magnets
- 25.** Sound takes some time to travel from one place to another. It will be minimum  
 (1) At night      (2) During summer      (3) During winter      (4) At evening
- 26.** The first planet of the solar system observed through the telescope was  
 (1) Mercury      (2) Saturn      (3) Uranus      (4) Neptune
- 27.** Lightning strike on earth occurs when buildings and ground are charged by the clouds. The charging process involved is

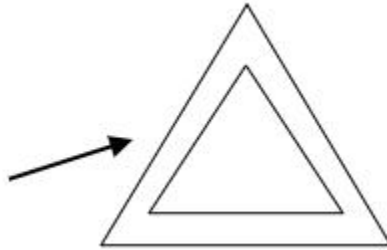


- (1) Friction      (2) Induction      (3) Conduction      (4) Earthing

**28.** During electrolysis of water, the gas evolved at the respective electrodes are

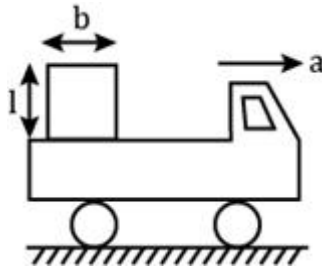
- (1) Anode - Oxygen, Cathode - Hydrogen      (2) Anode - Hydrogen, Cathode - Oxygen  
(3) Anode - Oxygen, Cathode - Oxygen      (4) Anode - Hydrogen, Cathode - Hydrogen

**29.** A ray of white light is incident on a hollow prism as shown. Then the light ray will undergo



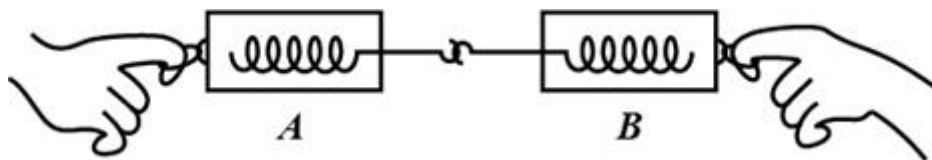
- (1) Change in its path and split into 7 colours  
(2) No change in its path but split into 7 colours  
(3) Change in its path but no splitting in 7 colours  
(4) No change in its path and no splitting in 7 colours

**30.** A box of dimensions  $L$  and  $b$  is kept on a moving truck which is accelerating towards right. If box does not slide on the truck, the force of friction acting on the block is



- (1) Towards left      (2) Towards right  
(3) At  $45^\circ$  towards left      (4) At  $45^\circ$  towards right

**31.** Consider two spring balances hooked as shown. We pull them in opposite directions. If the reading shown by A is 1.5 N, the reading shown by B will be



- (1) 1.5 N      (2) 2.5 N      (3) 3 N      (4) 0

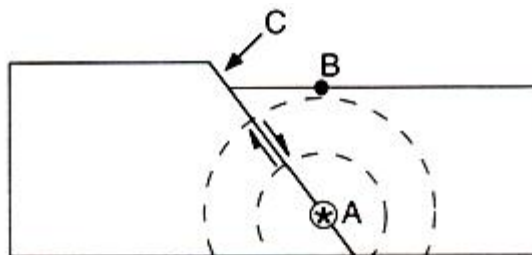
**32.** The distance between a compression and the next rarefaction in a longitudinal wave is

- (1)  $\lambda$       (2)  $\lambda/2$       (3)  $3\lambda/4$       (4)  $2\lambda$

**33.** In solar system, Pluto is not considered as a planet because

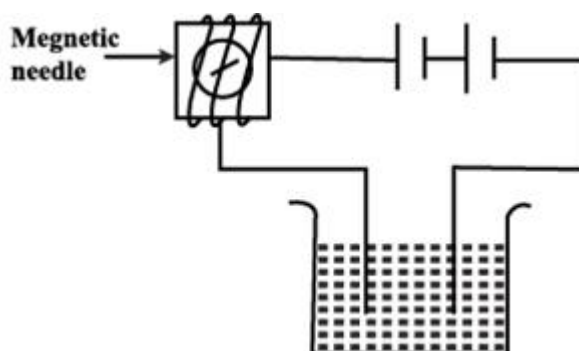
- (1) It is not in hydrostatic equilibrium      (2) It revolves very slow  
(3) It crosses the orbit of Neptune      (4) It does not revolve around Sun

34. Point A where slip initiated during earthquake is called



- (1) Dip (2) Epicenter (3) Focus (4) Scarp

35. Which liquid solution when tested in the manner shown causes the magnetic needle to deflect?



- (1) Glucose solution (2) Alcohol (3) Glycerine (4) Tap water

## SECTION-B : CHEMISTRY

This section contains **15 Multiple Choice Questions**. Each question has four choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

36. What is the full form of PET ?

- (1) Polyethylene terephthalate (2) Polyethylene triamine  
(3) Polyethylene tribasic (4) Polyethylene triacidic

37. The metal that reacts with cold water is :

- (1) Mercury (2) Sodium (3) Zinc (4) Bismuth

38. In an LPG cylinder, the gas is liquefied by

- (1) Increasing volume (2) Applying pressure  
(3) Increasing temperature (4) Decreasing pressure

39. Which of the following is closest to being an ideal fuel ?

- (1) Coal (2) LPG (3) Petrol (4) Kerosene

40. Carbon monoxide poisoning is most likely to occur in a \_\_\_\_.

- (1) open place with coal burning (2) room with LPG gas burning  
(3) closed room with coal burning (4) charcoal burning in an open space

41. Which of the following polymer is similar to cotton ?

- (1) Nylon (2) Dacron (3) Rayon (4) Terylene



- 42.** The tip of the lead pencil is made of  
(1) Lead (2) Graphite (3) Zinc (4) Charcoal
- 43.** The first scientist to sue the symbols of elements was  
(1) Dalton (2) Berzelius (3) Kanad (4) Proust
- 44.** The metal which does not react with air is :  
(1) Sodium (2) Calcium (3) Gold (4) Aluminium
- 45.** \_\_\_\_\_ coal has the lowest percentage of carbon.  
(1) Lignite (2) Bituminous (3) Anthracite (4) Peat
- 46.** CFCs are mainly responsible for  
(1) Creating smog (2) Creating a hole in ozone layer  
(3) Global warming (4) Alnino effect
- 47.** The oldest oil refinery in India is located at  
(1) Mathura (2) Digboi (3) Haldia (4) Kochi
- 48.** In which zone of a candle flame does complete combustion take place ?  
(1) Inner (2) Outer (3) Middle (4) All three zones
- 49.** A flame always points upwards because  
(1) gases produced in the flame are cold (2) gases produced in the flame are hot  
(3) hot gases are lighter and hence rises up (4) both (2) and (3)
- 50.** \_\_\_\_\_ is used as a fuel in rockets.  
(1) Liquid  $H_2$  (2) Liquid  $N_2$  (3)  $O_2$  (4) Liquid  $O_2$

### SECTION-C : BIOLOGY

This section contains **15 Multiple Choice Questions**. Each question has four choices (1), (2), (3) and (4) out of which ONLY ONE is correct.

- 51.** The endoplasmic reticulum in a cell probably functions in the  
(1) Protein synthesis (2) Transport of material  
(3) Both (1) and (2) (4) Secretory activities of cell
- 52.** Which of the following is not true for chemical fertilizers ?  
(1) They are nutrient specific (2) Soluble in water  
(3) Provide organic matter (humus) (4) Over use of fertilizers pollute the soil
- 53.** Penicillin is obtained from  
(1) Algae (2) Fungi (3) Lichens (4) Bacteria
- 54.** Kaziranga national park is famous for  
(1) Tiger (2) Musk deer  
(3) Elephant (4) One-horned rhinoceros

- 55.** During grafting the portion that is grafted is called as  
(1) Stock (2) Scion (3) Stalk (4) Stem
- 56.** Once a girl starts menstruating her body need more  
(1) iron (2) calcium (3) vitamin-C (4) vitamin-A
- 57.** The longest cell in human body is :  
(1) Neuron (2) Muscle fibre (3) Epithelial cell (4) Bone cell
- 58.** Which of the following crop would enrich soil with nitrogen ?  
(1) Apple (2) Pea (3) Paddy (4) Potato
- 59.** Which among the following is a connecting link between living and non-living ?  
(1) Bacilli (2) Yeast (3) Bacteriophages (4) Chlamydomonas
- 60.** Species found in a particular region are called as  
(1) Epidemic (2) Endemic (3) Threatened (4) Extinct
- 61.** Fertilization of ovum takes place in  
(1) Ovary (2) Fallopian tube (3) Cervix (4) Uterus
- 62.** In frog, the process of metamorphosis is regulated by  
(1) Insulin (2) Thyroxine (3) Estrogen (4) Progesteron
- 63.** Which among the following is/are viral disease ?  
(1) Chicken pox (2) Polio (3) Both (1) and (2) (4) Malaria
- 64.** Chloroplast are considered as self-replicating units as they contain  
(1) DNA (2) RNA (3) Both DNA & RNA (4) Lysosomes
- 65.** Forest Conservation Act was passed in  
(1) 1986 (2) 1980 (3) 1991 (4) 1972

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**SECTION-D : MATHEMATICS**

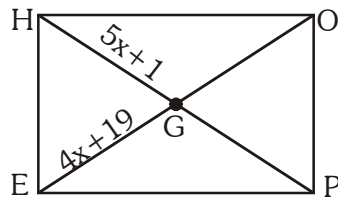
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This section contains **15 Multiple Choice Questions**. Each question has four choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

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- 66.** Which of the following is binomial ?  
(1)  $x^2 + x + 3$  (2)  $x^2 + 4$  (3)  $2x^2$  (4)  $x + 3 + \frac{1}{x}$
- 67.** The additive inverse of  $\frac{-a}{b}$  is  
(1)  $\frac{a}{b}$  (2)  $\frac{b}{a}$  (3)  $\frac{-b}{a}$  (4)  $\frac{-a}{b}$
- 68.** The range of the data 12, 25, 15, 18, 17, 20, 22, 6, 16, 11, 8, 19, 10, 30, 20, 32 is  
(1) 10 (2) 15 (3) 18 (4) 26

- 69.** A four digit number aabb is divisible by 55. Then possible value(s) of b is / are  
 (1) 0 and 2 (2) 2 and 5 (3) 0 and 5 (4) 7
- 70.** To construct a unique parallelogram, the number of measurements required is  
 (1) 2 (2) 3 (3) 4 (4) 5
- 71.** If the cost price of 10 shirts is equal to the selling price of 8 shirts, then which of the following is true for the transaction ?  
 (1) Profit of 25% (2) Loss of 25% (3) Profit of 20% (4) Loss of 20%
- 72.** Which of the following is not correct?  
 (1)  $\sqrt{0.4096} = 0.64$  (2)  $\sqrt{40.96} = 6.4$  (3)  $\sqrt{0.04096} = 0.064$  (4)  $\sqrt{4096} = 64$
- 73.** The solution of  $\frac{2x+3}{2x-1} = \frac{3x-1}{3x+1}$  is  
 (1)  $\frac{1}{8}$  (2)  $-\frac{1}{8}$  (3)  $\frac{8}{3}$  (4)  $-\frac{8}{3}$
- 74.** In a solid if  $F = V = 5$ , then the number of edges in this shape is  
 (1) 6 (2) 4 (3) 8 (4) 2
- 75.** In the figure, HOPE is rectangle. Its diagonals meet at G. If  $HG = 5x + 1$  and  $EG = 4x + 19$ , find x.



- (1) 15 (2) 18 (3) 20 (4) 16
- 76.** What is the area of the triangle ADE in the following figure.
- 
- (1)  $45 \text{ cm}^2$  (2)  $50 \text{ cm}^2$  (3)  $55 \text{ cm}^2$  (4)  $40 \text{ cm}^2$
- 77.** If  $\Rightarrow 9x^2 - \frac{1}{4} = 9x^2 - p$  then the value of p is :

- (1) 0 (2)  $-\frac{1}{4}$  (3)  $\frac{1}{4}$  (4)  $\frac{1}{2}$

**78.** If  $\frac{9^n \times 3^2 \times (3^{-n/2})^{-2} - (27)^n}{3^3 \times 2^3} = \frac{1}{27}$ . Then n =

(1) 0

(2) 1

(3) 2

(4) 3

**79.** Vishakha offers a discount of 20% on all the items at her shop and still makes a profit of 12%. What is the cost price of the article marked at Rs. 280 ?

(1) Rs. 100

(2) Rs. 150

(3) Rs. 200

(4) Rs. 300

**80.**  $\sqrt[3]{\sqrt[3]{a^3}} =$

(1) a

(2) 1

(3)  $a^{\frac{1}{3}}$

(4)  $a^3$

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SPACE FOR ROUGH WORK